Alberto Bucci

Curriculum Vitae

School of Mathematics
The University of Edinburgh, United Kingdom
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Current position

2025-now **Postdoctoral researcher**,
School of Mathematics, The University of Edinburgh, United Kingdom.

- **Research project**: Randomized Numerical Linear Algebra for Optimization and Control of PDEs.
- **Principal investigators**: Prof. John Pearson and Prof. Stefan Güttel.

Previous positions

2024-2025 **Postdoctoral researcher**,
Department of Numerical Mathematics, Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic.

- **Research project**: A Lanczos-like Method for the Time-Ordered Exponential.
- **Principal investigator**: Prof. Stefano Pozza.

2021-2024 **PhD student in Mathematics**,
Department of Mathematics, University of Pisa, Italy.

- **Supervisor**: Prof. Leonardo Robol.
- **Research area**: Randomized techniques for the low-rank decomposition of tensors.
- **PhD defense date**: 24 February 2025.
- **Final mark**: Ottimo con lode (highest grade with honors, Italian system).

2020-2021 **Collaboration contract**,
Department of Mathematics, Scuola Normale Superiore, Italy.

- **Coordinators**: Prof. Michele Benzi & Prof. Fabrizio Lillo.
- **Project description**: Development of a Python library for describing, modelling, and studying complex and dynamic networks: NetworkSNS.

Education

2017-2019 **Master degree in Mathematics**,
Department of Mathematics, University of Pisa, Italy.

- **Thesis**: Newton-type methods for the Multilinear PageRank.
- **Advisor**: Prof. Federico Poloni.
- **Date**: December 13, 2019.
- **Final mark**: 110/110.

2014-2017 **Bachelor degree in Mathematics**,
Department of Mathematics, University of Pisa, Italy.

- **Thesis**: Post-quantum cryptography: NTRU.
- **Advisor**: Prof. Carlo Traverso.
- **Date**: September 23, 2017.
- **Final mark**: 110/110 cum Laude.

Teaching

2023-2024 Tutor in Scientific Calculus (University of Pisa).

2022-2023 Tutor in Scientific Calculus (University of Pisa).

2021-2022 Tutor in Laboratory of Computational Mathematics (University of Pisa).

2021 Supply Maths Teacher (Penne, Italy).

2015-2016 Math Olympiad Lecturer for the Italian Mathematical Union (Bologna, Italy).

Thesis supervision

2026 **MSc dissertation**, MSc in Computational Applied Mathematics,
School of Mathematics, University of Edinburgh, UK.

- **Title**: Low-Rank Tensor Decomposition for the Solution of Partial Differential Equations.
- **Student**: Jinjiang Hu.
- **Role**: Supervisor.

2026 **Bachelor thesis in Mathematics**,
Department of Mathematics, University of Pisa, Italy.

- **Title**: Robust Principal Component Analysis: teoria, algoritmi e applicazioni all'elaborazione video (*Robust Principal Component Analysis: theory, algorithms and applications to video processing*).
- **Student**: Luca Matteoni.
- **Role**: Co-supervisor, with Prof. Leonardo Robol.

Publications

2026 **Fast matrix-vector products in tensor-train format via structured sketching.**
arXiv:2610.23451
A.B., S. Güttel, J. W. Pearson

2026 **Randomized Tucker-sketched GMRES.**
arXiv:2608.11091
A.B., M. Iannacito, M. Pasha, R. Smith

2026 **An equivalence result for sketched Anderson acceleration and sketched GMRES.**
arXiv:2607.04904
A.B., F. Durastante

2026 **Randomized flexible LSQR and LSMR with applications to inverse problems.**
arXiv:2605.22281
A.B., S. Gazzola, L. Robol

2026 **Randomized algorithms for streaming low-rank approximation in tree tensor network format.**
Numerical Linear Algebra with Applications — doi.org/10.1002/nla.70107
A.B., G. Verzella

2025 **Numerical stability of the Nyström method.**
arXiv:2511.15583
A.B., Y. Nakatsukasa, T. Park

2025 **Randomized sketched TT-GMRES for linear systems with tensor structure.**
SIAM Journal on Scientific Computing — doi.org/10.1137/24M1694999
A.B., D. Palitta, L. Robol

2025 **A sequential multilinear Nyström algorithm for streaming low-rank approximation of tensors in Tucker format.**
Applied Mathematics Letters — doi.org/10.1016/j.aml.2024.109271
A.B., B. Hashemi

2024 **A multilinear Nyström algorithm for low-rank approximation of tensors in Tucker format.**
SIAM Journal on Matrix Analysis and Applications — doi.org/10.1137/23M1599343
A.B., L. Robol

2022 **A continuation method for computing the multilinear PageRank.**
Numerical Linear Algebra with Applications — doi.org/10.1002/nla.2432
A.B., F. Poloni

Research visits

2026 **ICERM, Brown University**,
Providence, Rhode Island (90 days).

- Participation in the semester program "Stochastic and Randomized Algorithms in Scientific Computing: Foundations and Applications".

2024 **University of Oxford**,
Oxford, United Kingdom (40 days).

- Visit to Prof. Yuji Nakatsukasa to start a now finalized collaboration, which later led to a collaboration on the stability of Nyström approximation.

International collaborators

- Behnam Hashemi, University of Leicester, United Kingdom.
- Yuji Nakatsukasa, University of Oxford, United Kingdom.
- Taejun Park, EPFL, Switzerland.
- Gianfranco Verzella, University of Geneva, Switzerland.
- Stefano Pozza, Charles University, Czech Republic.
- Rudi Smith, Virginia Tech, Virginia.
- Yiyun He, University of California, California.
- Swati Padmanabhan, University of Minnesota Twin Cities, Minnesota.
- Arvind K. Saibaba, North Carolina State University, North Carolina.

Other scientific activities

Referee for the following journals

- SIAM Journal on Matrix Analysis and Applications
- Applied Mathematics and Computation
- Applied Numerical Mathematics
- Linear Algebra and Its Applications
- Linear and Multilinear Algebra
- Numerical Algorithms
- Numerical Linear Algebra with Applications
- Operators and Matrices

Organizer of the following seminars



Founder and organizer (2022-2024) of Pysanum (Pisan Young Seminars in Applied and NUmerical Mathematics): a seminar series on numerical analysis and applied mathematics tailored for students and young researchers (now also a SIAM Chapter).

Schools

- Participation in the summer school "Gene Golub SIAM Summer School on Iterative and Randomized Methods for Large-Scale Inverse Problems" (Quito, Ecuador, 2024).
- Participation in the summer school "High-Dimensional Approximation: From Theoretical Foundations to Machine Learning and PDEs" (Cetraro, Italy, 2024).
- Participation in the summer school "Machine Learning: From Data to Mathematical Understanding" (Cetraro, Italy, 2023).
- Participation in the Mini Course on Computational Methods for Large-Scale Matrix Equations and Application to PDEs. Professor: Valeria Simoncini (L'Aquila, Italy, 2021).
- Participation in the winter school "Fourth EACA International School on Computer Algebra and its Applications" (Santiago De Compostela, Spain, 2018).

Invited talks

- 27th Conference of ILAS (Blacksburg, Virginia, May 18-22, 2026).
- ICERM workshop on Randomized Numerical Linear Algebra (Providence, Rhode Island, February 2-6, 2026).
- Basics, Signal treatment, Simulation, Numerical mathematics (Paris, France, July 16-17, 2025).
- Pisan Young Seminars in Applied and NUmerical mathematics (Pisa, Italy, September 29, 2025).
- SIAM Conference on Applied Linear Algebra (Paris, France, May 13-17, 2024).
- Sketching, Mixed Precision, and Associated Algorithms for Scientific Computing (Bologna, Italy, January 18-19, 2024).
- 25th Conference of ILAS (Madrid, Spain, June 12-16, 2023).

Contributed talks

- Householder Symposium (Ithaca, New York, June 23-28, 2025).
- Biennial Numerical Analysis Conference (Glasgow, United Kingdom, June 10-11, 2025).
- Joint GNCS-SIAM Chapters Meeting for Young Researchers in Numerical Analysis and Applied Mathematics (Pavia, Italy, February 10-11, 2025).
- METT X - 10th Workshop on Matrix Equations and Tensor Techniques (Aachen, Germany, September 13-15, 2023).
- Numerical Linear Algebra Days (L'Aquila, Italy, May 10-12, 2023).
- GAMM ANLA Workshop (Prague, Czech Republic, September 22-23, 2022).

Other conferences

- Exploiting Algebraic and Geometric Structure in Time-Integration Methods (Pisa, Italy, May 3-5, 2024).
- Due Giorni di Algebra Lineare Numerica e Applicazioni (Napoli, Italy, February 14-15, 2022).
- Due Giorni di Algebra Lineare Numerica (Rome, Italy, February 18-19, 2019).

Projects and grants

2026 Research project GNCS: Operatori non locali e solutori efficienti per Sistemi COmplessi in MOdelli continui e DIscreti.

2024 Research project GNCS: Metodi di riduzione di modello ed approssimazioni di rango basso per problemi alto-dimensionali.

2023 Research project GNCS: Metodi basati su matrici e tensori strutturati per problemi di algebra lineare di grandi dimensioni.

2022 PRIN 2022 Project: Low-rank Structures and Numerical Methods in Matrix and Tensor Computations and their Application.

Awards

2017-2019 INDAM Mathematical research fellowship – Istituto Nazionale di Alta Matematica "Francesco Severi".

2014-2017 INDAM Scholarship for the enrollment at maths – Istituto Nazionale di Alta Matematica "Francesco Severi".

2014 Finalist at the Italian Physics Olympiad.

2014 Finalist at the Italian Mathematics Olympiad.

Foreign languages

Italian Mother tongue

English Fluent speaking and writing skills (Certificate: B2 First – Cambridge Assessment English)

Research interests

The main focuses of my research are randomized techniques and tensors, with a particular focus on low-rank decomposition and resolution of large-scale linear systems. I have also worked on sparse matrix computations, iterative methods, and matrix functions with applications to complex networks and Markov chains. Currently, I am also interested in applications in quantum computing and AI.